Moses Kwabla Alorse

Email: mkalorse@gmail.com Phone: +1-437-665-2045
LinkedIn: linkedin.com/in/alorse-moses-698a4569

Profile Summary

Quantitative Economist with double Master's degrees in Economics and Mathematics and advanced training in econometrics, policy evaluation, financial modeling, and statistical programming. Skilled in volatility forecasting, asset pricing analysis, and portfolio optimization using Python and R. Adept at translating complex financial data into actionable insights for risk management and investment decision-making. Proficient in developing and simulating dynamic growth models in MATLAB to evaluate the fiscal and macroeconomic impacts of economic policy interventions. Experienced in causal inference modeling, including potential outcomes framework and Difference-in-Differences (DiD) designs, to evaluate treatment effects while controlling for covariates using R.

Education

Ph.D. in Economics Sept. 2024 – Present
York University, Toronto, Canada
MA in Economics Sept. 2023 – Jun. 2024
York University, Toronto, Canada
Post-Graduate Diploma in Education Aug. 2019 – Aug. 2020
University of Education, Winneba, Ghana
MSc. in Mathematics Aug. 2015 – Jun. 2016
University of Cape Coast, Cape Coast, Ghana
BA in Economics and Mathematics Aug. 2009 – May 2013
University of Ghana, Legon, Ghana

Skills and Competencies

Programming: Python (NumPy, SciPY, Pandas, Matplotlib, statsmodels), SPSS, EViews, R, MATLAB, Stata
Quant Methods: Econometrics, Financial Modeling (ARMA, GARCH), Panel Data, MLE, Constraint Optimization
Modeling: SDGE, RBC, CAPM, Bayesian Games, DiD, Potential Outcomes
Tools: LaTeX, Advanced Excel, SQL (intermediate)
Soft Skills: Teaching, Teamwork, Communication, Time Management, Independent Work, Multi-tasking.

Selected Quantitative Projects at York University

Volatility Modeling of Equity Returns

- Modeled daily Apple stock data (Jan. 2019 – Dec. 2022) using ARMA and GARCH in R to assess risk and return patterns and forecast volatility.
- Evaluated out-of-sample volatility forecasts and Value-at-Risk performance.
- Estimated ARMA-GARCH models in R to forecast volatility of Canadian equities, improving short-term risk prediction accuracy.

Stock Performance and Random Walk Testing

- Tested the performance of 30 US stocks using Random Walk One and Random Walk Three models in R and advanced Excel to evaluate market behavior and predictability.
- Implemented Wald tests and cross-sectional regression analysis of 30 US stocks using CAPM, implemented in Excel and R, to evaluate risk-return relationships and asset performance.
- Constructed optimal portfolio weights and the efficient frontier for 30 US stocks using Markowitz (1952) methodology, employing the S&P 500 as a market proxy, implemented in R.

Dynamic Consumption-Growth Simulation and Economic Policy

- Analyzed US Labor Market CPS data using advanced probability models in R Markdown and Stata to assess the impact of demographics on income distribution and employee job stability.
- Simulated growth and consumption paths in stochastic dynamic models using MATLAB.
- Conducted causal inference analysis using panel data methods in Stata, applying fixed effects and difference-in-differences models to estimate treatment effects.

Work Experience

Graduate Assistant Sept.2023– Dec.2025
York University, Toronto

- Explained advanced economic theories and concepts to students, improving understanding and learning outcomes.
- Collaborated with faculty members to deliver tutorials and reinforce core economic concepts, including microeconomics, macroeconomics, and quantitative methods.
- Guided students in solving problem sets involving optimization, equilibrium analysis, regression models, and statistical inference.
- Provided individualized academic support during office hours, fostering an inclusive and supportive learning environment.
- Communicated effectively with diverse student groups, enhancing inclusive engagement and academic performance.

Teaching Experience

Teacher Jan. 2017 – Aug. 2023
Ghana Education Service(GES), Ghana

- Improved final examination pass rate from 60% to 92% in 2018, and sustained consistent high performance of 70–90% between 2019 and 2022 through targeted mentoring and curriculum support.
- Organized and led supplementary teaching sessions for Science and Mathematics Club students, resulting in their first quarter-final placement in the Regional Science and Mathematics Quiz Competition.
- Designed and implemented a structured election process, improving transparency and operational efficiency.
- Coordinated and managed the girls' soccer team (2019–2022), overseeing training schedules, competition logistics, and performance tracking to ensure successful participation in zonal and inter-school tournaments.

Teaching Assistant Sept. 2013 – Aug. 2014
Mathematics Department – University of Ghana, Legon

- Explained advanced mathematical and statistical theories and concepts to students, improving understanding and learning outcomes.
- Assisted in grading and evaluating complex mathematical and statistical problem sets, demonstrating strong quantitative reasoning.

Extracurricular Activities and Additional Information

Player and Assistant Coach, Ave – Teachers Soccer Team Dec. 2017 – Aug. 2023

- Assistant Coach and Player, Teachers Soccer Team (2017–2023).
- Languages: English (Fluent).
- Interests: Data Analytics, Quantitative Research, Programming.